

Evaluating Offline Reinforcement Learning for Sepsis

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Abstract. In the ICU, sepsis is one of the leading causes of death. Vaso-pressor and fluid dosing in sepsis is still not standardised, despite decades of clinical research, since the right dose depends on how a patient’s condition evolves and not on a fixed protocol. This is fundamentally a sequential decision problem, one that offline reinforcement learning can potentially address by learning treatment policies directly from historical ICU trajectories, without exposing any patient to experimental dosing. Prior work in this area rests on off-policy evaluation (OPE), where an estimator scores a learned policy using trajectories collected under the behaviour policy. Such estimators either reweight the clinician’s trajectories through importance sampling or fit a value function directly, and neither protocol is unconditionally reliable for this setting. Because a learned policy is deterministic, importance sampling’s weight collapses to zero the moment it recommends an action the clinician did not take, shrinking the effective sample size, while fitted Q evaluation avoids this by bootstrapping instead, at the cost of extrapolating beyond the support of the behaviour policy. Yet existing studies often report these estimates as results without testing whether the estimators are reliable on the data at hand.

We treat the evaluation itself as the object of study. Three RL models (D3QN, CQL and IQL) are trained on a Sepsis-3 cohort of MIMIC-IV stays and scored in two independent ways: an off-policy value estimate, computed with four estimators (weighted and per-decision importance sampling, doubly robust and fitted Q evaluation), and state-action coverage, which measures how much of a policy’s recommended actions the data has evidence for.

Keywords: Offline reinforcement learning · Sepsis · MIMIC-IV · Off-policy evaluation · Q-value extrapolation · Data support · Conservative Q-Learning

1 Introduction

Sepsis is a life-threatening organ dysfunction, caused by a dysregulated host response to infection. A landmark analysis of the Global Burden of Disease

Study estimated 48.9 million sepsis cases worldwide in 2017, with 11 million sepsis-related deaths that year alone: roughly one in five deaths globally (Rudd et al., 2020; Singer et al., 2016). Mortality remains high even in well-resourced ICUs: a quarter of the MIMIC-IV sepsis patients studied here die within 90 days. Despite decades of clinical trials, no single treatment protocol works best for all patients, and ICU clinicians managing sepsis must revisit their treatment decisions every few hours, adjusting vasopressor doses and intravenous (IV) fluid volumes as the patient’s condition evolves. The 2021 Surviving Sepsis Campaign guidelines (Evans et al., 2021) give a broad framework, yet dosing remains hard to standardise.

Reinforcement learning (RL) suits this kind of sequential decision problem well, because it optimises directly for long-term outcomes and not for short-term predictions. Komorowski, Celi, Badawi, Gordon, and Faisal (2018) showed with the “AI Clinician” that patients whose treatment matched an RL-derived policy most closely had lower 90-day mortality in MIMIC-III. Roggeveen et al. (2021) built on this with a Dueling Double DQN (D3QN) and a SOFA-augmented reward, getting cleaner policy convergence on the same cohort.

Healthcare RL faces a hard constraint: the agent can only learn from historical records, since deploying an untested policy on real patients is not an option. Thus, in a clinical setting, we cannot use online RL. This offline batch setting causes *distributional shift*: the policy may assign high values to actions clinicians rarely took, making its Q-value estimates unreliable (Levine, Kumar, Tucker, & Fu, 2020). Evaluation is equally difficult. Without deployment, we must rely on *off-policy evaluation* (OPE) to estimate policy value from historical data. As we show, three of the four estimators fail on ICU trajectories. The importance-sampling estimators end up resting on a handful of patients, and the doubly robust estimator yields values no policy could achieve. Fitted-Q evaluation alone returns a usable value for every policy, including those whose recommendations the data never recorded.

These two difficulties compound in a way that can easily be missed. An agent that exploits distributional shift is assigned a high value by its own network, and an estimator fitted on the same records extrapolates over the same actions. Published comparisons rarely surface this, since an improvement usually rests on a single architecture scored by a single estimator, which leaves a genuine gain hard to separate from a shared extrapolation error. This work treats the evaluation itself as the object of study: we establish which estimator can carry a claim on sequential ICU data, and only then compare the policies those algorithms produce. Estimator reliability has been studied before, though mostly on simulation benchmarks rather than clinical data (Voloshin, Le, Jiang, & Yue, 2021), and Sec. 2 addresses what that literature does and does not settle for trajectories of ICU length.

We argue that two different measurements describe the quality of a treatment policy learned offline: the value an estimator assigns to it, and how much clinical practice in the data stands behind its recommendations. On this cohort, the two tend to diverge. To establish this claim we set up an evaluation protocol. We

assess the reliability of four off-policy estimators on ICU treatment trajectories, including failure modes prior work has largely overlooked. We build a data-support diagnostic that pairs a propensity measure with a k -nearest-neighbour (k -NN) measure, recording how far each policy’s recommendations are covered by observed practice. We measure the estimator’s own variance, so that a reported margin can be told apart from noise. Under that protocol, we then compare unconstrained, implicitly conservative and explicitly conservative agents.

This gives one overarching research question:

How reliably can offline reinforcement learning policies for sepsis treatment be evaluated?

It divides into three parts, which structure the results:

- RQ1** Which off-policy estimators (WIS, PDWIS, DR, FQE) remain usable on ICU treatment trajectories, and what claims can their output support?
- RQ2** Where an estimator is defined, how much of its estimate rests on treatments clinicians actually gave?
- RQ3** Do estimated value and data support identify the same policies, and does any policy satisfy both?

2 Related Work

RL for Sepsis Treatment. Raghu et al. (2017) began this line of work, applying deep Q-learning to vasopressor and fluid management on MIMIC-III, and estimated that following the learned policy would lower hospital mortality by up to 3.6% against a 13.7% baseline. The approach became influential with the “AI Clinician” of Komorowski et al. (2018), which grouped similar patients into clusters and split treatment into a 5×5 grid of vasopressor and fluid doses. Two encouraging results came out of this experiment. The learned policy scored higher than the clinicians, and mortality was lowest among patients whose real treatment happened to sit closest to what the policy recommended. It is that second result that the field has been trying to reproduce or refute ever since. Peng et al. (2018) combined a kernel-based learner with a deep Q-network in a mixture of experts, switching by local data density. Closest to our own work is Roggeveen et al. (2021), who used a D3QN and added a reward term based on the SOFA organ-failure score, which made training converge more cleanly. We build on its network, though not on its reward shaping. Choi et al. (2024) train a similar network on the same 25-action grid and report up to 10% higher estimated survival on MIMIC-III and eICU. That number comes from dose-matching: patients are grouped by how close their real treatment was to the policy’s recommendation, and the survival of the best-matching group is read as the policy’s value. Dose-matching is the most optimistic of the estimators in use, and the least often checked. Choi et al. (2024) note it does not give a true survival rate.

Offline RL and Distributional Shift. Standard DQN (Mnih et al., 2015) trained offline tends to overestimate Q-values for actions outside the training data. This problem is known as Q-value extrapolation (Fujimoto, Meger, & Precup, 2019). It is not an artefact of deep networks: classical fitted-Q iteration (Ernst, Geurts, & Wehenkel, 2005) maximises over the same unsupported actions and inherits the same instability offline. Conservative Q-Learning (CQL) (Kumar, Zhou, Tucker, & Levine, 2020) fixes this by penalising Q-values for unsupported actions. Batch-Constrained Q-learning (BCQ) (Fujimoto et al., 2019) instead fits a generative model of the behaviour policy and restricts the policy to actions that model rates as plausible, so an action the clinicians never took is never evaluated. More recently, distributional RL (Bellemare, Dabney, & Munos, 2017) models the full return distribution, not just its average. This gives richer uncertainty estimates, and potentially more cautious policies offline (Roggeveen et al., 2024).

Off-Policy Evaluation. Evaluating RL policies without deploying them matters most in clinical settings. Importance sampling (IS) estimators are unbiased on average, but the variance of the trajectory-wise weight grows as a power of the trajectory length, with a base set by the divergence between the evaluated and behaviour policies (Liu, Li, Tang, & Zhou, 2018). A deterministic policy drives that base up, so a single action mismatch anywhere in a stay removes it from the estimate, which Raghu et al. (2017) found on sepsis data and decided to fall back on a model-based estimate. Per-decision WIS (PDWIS) (Precup, Sutton, & Singh, 2000) reduces that variance by normalising weights at each time step instead of over the whole trajectory. The Doubly Robust (DR) estimator (Jiang & Li, 2016) combines a model-based value estimate with an IS correction, giving lower variance when the value model fits well, and inheriting that model’s errors when it does not. Fitted Q Evaluation (FQE) (Le Paine et al., 2020) trains a separate Q-network to estimate a policy’s value from offline data, avoiding importance sampling altogether. Tang and Wiens (2021) showed the practical risk of OPE-based model selection in healthcare: a policy chosen on one validation set may not generalise. Voloshin et al. (2021) ran a large empirical comparison of OPE methods. They found FQE competitive with, or even better than, IS-based alternatives across eight simulation benchmarks, none of them clinical and none with trajectories of ICU length.

Two features of this literature motivate the present work. First, reported improvements are almost always established under a single estimator, and frequently under one whose reliability in this setting has not been checked. The studies above rely on several different estimators: weighted importance sampling, doubly robust, weighted doubly robust and dose-matching. The reported margins are small relative to the spread those estimators introduce. In Peng et al. (2018), changing which value function the weighted doubly robust estimator takes as its control variate moves the mixture policy’s score by more than the deep network’s whole improvement over the clinicians. Second, these studies do not report the coverage behind a value estimate. Where coverage is handled at all it is built into the learning algorithm, as in the conservative penalty of Kumar et

al. (2020), rather than measured afterwards. Nauka et al. (2025) show what the omission costs. Applying the AI Clinician to emergency-department care, they find that performance degrades under missing features and shifted treatment patterns, and conclude that generalisation more than in-distribution accuracy is the field’s open problem. We therefore take the evaluation as the object of study. We test which estimators work on these trajectories and establish a diagnostic for the support behind their estimates.

3 Methodology

This section states what each component is and why it was chosen.

3.1 MDP Formulation

We model sepsis treatment as a finite-horizon discrete-action Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S} \subseteq \mathbb{R}^{42}$ is the space of patient states, $\mathcal{A}$ the discrete treatment grid, $P(s_{t+1} | s_t, a_t)$ the unknown transition dynamics of the patient (never modelled explicitly, since all methods here are model-free), R the reward defined below, and $\gamma = 0.99$ the discount factor, high enough that the terminal outcome still reaches the first decision of the longest episode at $0.99^{17} = 0.84$. Time is discretised into 4-hour bins anchored at sepsis onset, giving a median horizon of ten steps and a maximum of eighteen. At each step the agent observes s_t and selects a_t , with the objective of maximising $\mathbb{E}[\sum_t \gamma^t r_t]$. The 4-hour resolution is inherited from Komorowski et al. (2018) and retained here because finer bins worsen missingness (most laboratory values are drawn far less often than hourly) while lengthening the horizon, which degrades both importance sampling and fitted-Q evaluation.

State space. The state $s_t \in \mathbb{R}^{42}$ has four parts. The first is 7 vital signs: heart rate, systolic/diastolic/mean arterial blood pressure, respiratory rate, SpO₂, and temperature. The second is 24 laboratory values: lactate, WBC, platelets, creatinine, bilirubin, haemoglobin, BUN, electrolytes, coagulation, blood gases, bands and CRP. The third is 5 derived or contextual measures: FiO₂, GCS total, the SOFA total, an indicator of whether that total was observed at the boundary or carried forward, and urine output. The fourth records treatment context and demographics: the vasopressor and fluid levels given in the preceding interval, age, biological sex, hours since ICU admission, and hours from sepsis onset. The representation itself is continuous and unclustered, following Raghu et al. (2017) rather than the k-means clustering of Komorowski et al. (2018). Its physiological composition, however, follows Komorowski et al. (2018) and Roggeveen et al. (2021), whose variable sets we adopt so that the state is comparable with prior sepsis-RL work.

One inclusion differs from those two templates. The previous interval’s treatment levels enter the state because the reward is terminal and the action space

is a dose grid. Without them the network would not be able to tell a patient who is stable on support from one who is stable without it.

Action space. The action space $\mathcal{A}$ is a grid over vasopressor intensity and IV fluid volume, following the discretisation used by Komorowski et al. (2018) and Roggeveen et al. (2021), whose bin counts we adopt. Vasopressor doses are first converted to norepinephrine equivalents (mcg/kg/min) using standard conversion factors (Brown et al., 2013). Each axis has five levels. Level 0 is an explicit test for zero, so it denotes no drug and no fluid given in the interval. The four remaining levels are divided at the quartiles of the non-zero doses, learned on the training split alone so that no held-out patient informs the discretisation. That places the vasopressor cut-points at 0.060, 0.114 and 0.261 mcg/kg/min and the fluid cut-points at 40, 213 and 501 mL per four hours. Learning the edges from the data, instead of fixing them in advance, keeps all 25 actions reachable and populated, and the zero test separates withholding a treatment from giving a small amount of it, which is the distinction the comparison against a never-treat control depends on. Under this grid 16.8% of clinician decisions give neither drug nor fluid.

Reward. The outcome that matters is only known once the stay is over, so the reward carries that verdict and nothing else. It is terminal and sparse:

$$r_t = 15 \cdot o_t \cdot d_t, \quad (1)$$

where d_t is a terminal indicator and $o_t \in \{+1, -1\}$ records survival to 90 days after ICU admission, all-cause, following the sepsis-RL literature (Komorowski et al., 2018). Every non-terminal reward is exactly zero. The episode ends 48 hours after sepsis onset, while survival is measured at 90 days from ICU admission, so the reward covers a long period in which the agent takes no actions.

A shaping term proportional to the change in SOFA, as used by Roggeveen et al. (2021), is deliberately not added. It imports a proxy objective, so a policy can be rewarded for moving a score without changing survival, and the reason usually given for it does not apply here: episodes are capped at 18 intervals, so the discounted terminal signal still reaches the first decision at $0.99^{17} = 0.84$ and there is little horizon left for shaping to bridge.

The sparse form also fixes the scale. Every trajectory carries exactly one non-zero reward, +15 or −15, so no state has a true value outside that range under any policy. A Q -value above it cannot have come from the data and can only come from a network generalising past the actions the data contains, so the excess measures extrapolation and not value.

Episodes. One episode spans from −24 hours to +48 hours after sepsis onset. Making one episode last 72 hours, with 18 timesteps (4 hours being one timestep). Each onset-centred window constitutes one episode, referred to as a trajectory when discussing the state–action–reward sequence it produces. The terminal transition is the final 4-hour bin of the window, clipped to the stay, and the outcome reward is assigned at that step.

3.2 Cohort and Dataset

We use MIMIC-IV (v3.1) (Johnson et al., 2023), a public critical-care database. Sepsis is identified by the Sepsis-3 criteria of Singer et al. (2016), namely suspected infection together with an acute rise of at least two points in the SOFA score in the surrounding window. We take both the classification and the onset timestamps from the derived Sepsis-3 table shipped with MIMIC-IV, instead of reimplementing the criteria, so the cohort is the one the database maintainers define and is reproducible without reference to our own code. That table holds 32,899 qualifying stays.

Two filters apply. Patients must be at least 18 years old, and each patient contributes only their first eligible episode, so that no patient enters the cohort more than once. This leaves 25,442 stays. There is deliberately no length-of-stay filter. Prior sepsis-RL work commonly requires a stay between one and thirty days, but the unit of analysis here is an episode anchored to sepsis onset and not a whole admission, so a valid episode can sit inside a stay shorter or longer than those bounds. Filtering on the stay would discard episodes on a criterion unrelated to whether they contain a treatment sequence.

Anchoring on onset makes the population comparable across patients, because every trajectory begins at the same point in the illness and a state at the third interval therefore means the same thing for everyone.

Ninety-day all-cause mortality in this cohort is 25.0%, against roughly 20% in the unfiltered ICU population. A sepsis criterion is expected to produce that separation, which is consistent with the cohort not being simply a general critical-care population.

Stays are split by subject and not by time, so every transition from one patient falls into the same partition and no patient appears in both training and evaluation, precisely to avoid patient-level data leakage.

3.3 Q-Network Architecture

All three methods share one Q-network, a Dueling Double Deep Q-Network (D3QN), following Roggeveen et al. (2021) and Wang et al. (2016). The two qualifiers name two separate departures from a standard DQN. *Double* learning uses the online network to choose the best next action but the older target network to score it, which curbs the tendency of a single network to favour whichever action it has already overrated. *Dueling* splits the output into two heads: one estimates how good the state itself is, the other how much each action adds on top. That split suits this problem, because the patient’s condition drives most of the variation between time bins, and the network can learn that a patient is deteriorating without having seen every dose tried on similar patients.

The two heads are recombined in the standard dueling form,

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a' \in \mathcal{A}} A(s, a'), \quad (2)$$

where $V(s)$ is the value of the state on its own, $A(s, a)$ the advantage of action a over that baseline, and $\mathcal{A}$ the action space.

The width and depth are taken from Roggeveen et al. (2021) and not tuned here, for the same comparability reason as the state and reward. At about 85k parameters the network is small relative to the 176,012 training transitions, so capacity is not the binding constraint. The failure mode is extrapolation. We train it offline with the Double DQN target (van Hasselt, Guez, & Silver, 2016), which limits the maximisation bias that offline bootstrapping amplifies, and with the Huber loss, which bounds the gradient contribution of the large temporal-difference errors that Q -value extrapolation produces.

3.4 Model Variants

The three methods compared share that network and differ only in how they treat out-of-distribution actions. Presenting them together makes that axis explicit, since it is the axis along which the results separate.

Conservative Q-Learning (CQL). CQL (Kumar et al., 2020) attacks the extrapolation problem directly, adding a term to the Bellman (Huber) loss that lowers Q -values on out-of-distribution actions while raising the value of the action actually taken in the data:

$$\mathcal{L} = \mathcal{L}_{\text{Bellman}} + \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp Q(s, a) - Q(s, a_{\text{data}}) \right]. \quad (3)$$

The bracketed penalty is smallest when probability mass concentrates on the dataset action a_{data} . The coefficient $\alpha \geq 0$ sets the strength of conservatism ($\alpha = 0$ recovers plain D3QN). The architecture, optimiser and all other hyperparameters match D3QN exactly, so the penalty is the only difference between the two.

Implicit Q-Learning (IQL). IQL (Kostrikov, Nair, & Levine, 2022) suppresses the same extrapolation without an explicit penalty, and is included to separate the two mechanisms. The Bellman target of a Q -network ordinarily takes a maximum over all actions, which is where an unsupported action enters. IQL replaces that maximum with a separate state-value network $V(s)$, trained by expectile regression to sit near the upper end of the Q -values of actions that appear in the dataset $\mathcal{D}$,

$$\mathcal{L}_V = \mathbb{E}_{(s,a) \sim \mathcal{D}} [\ell_\tau^2(Q(s, a) - V(s))], \quad (4)$$

where $\ell_\tau^2(u) = |\tau - \mathbb{I}[u < 0]| u^2$ and $\tau \in (0.5, 1)$ sets how close to the maximum V is pushed. Q is then regressed toward $r + \gamma(1 - d)V(s')$. No loss evaluates Q at an action the data does not contain, so extrapolation never enters the bootstrap, though unlike CQL the out-of-distribution Q -values themselves are left untouched. The critic, meaning the Q -network, reuses the D3QN architecture. Once it is trained it is frozen, and a separate actor network is extracted from it by advantage-weighted behaviour cloning. That actor is the policy evaluated here, not a softmax over the critic’s Q -values.

3.5 Off-Policy Evaluation

The value of each policy needs to be estimated from logged data in OPE. We run four estimators instead of one, and the reason is central to this thesis: no off-policy estimator is unconditionally reliable, and each rests on a different assumption, so a single estimator cannot separate a failure of the policy from a failure of the measurement. The four span the two available families. WIS and PDWIS are importance-sampling estimators, which need a correct behaviour policy but no value model. FQE is a model-based estimator, which needs a value model but no importance weights. DR combines both: it is unbiased if either ingredient is correct, but its variance is carried by the importance weights, so it is exposed to the same collapse as WIS. Sec. 5.1 reports which of the four remain usable on this data.

Per-decision Weighted Importance Sampling (PDWIS). Importance sampling reweights the clinician’s logged trajectories to stand in for ones the evaluated policy would have produced. Writing π_e for the policy under evaluation and T for the last step of the stay, each step contributes a ratio $\pi_e(a_t|s_t)/\pi_\beta(a_t|s_t)$, large where the policy would have picked the action the clinician actually chose and small where it would not. The trajectory-level WIS estimator multiplies all of them into one weight $\rho_{0:T} = \prod_{t=0}^T \pi_e(a_t|s_t)/\pi_\beta(a_t|s_t)$ and takes the weighted average discounted return. A single action mismatch anywhere in a stay drives that product to zero, so the average is carried by the few trajectories that agree throughout. How few is measured by the effective sample size, $(\sum_i w_i)^2 / \sum_i w_i^2$, the number of equally weighted trajectories that would give the same variance. PDWIS (Precup et al., 2000) normalises the weights at each step instead of over the whole trajectory, which bounds the damage a single mismatch can do.

Doubly Robust (DR). DR (Jiang & Li, 2016) begins from the value the Q-network predicts, then uses importance weighting to correct that prediction where the logged data departs from it. The name refers to its guarantee: the estimate stays unbiased if *either* the behaviour policy or the value model is correct, without requiring both. Jiang and Li (2016) show its variance falls below PDWIS’s when the value model fits well, and we use their formulation unchanged. Written recursively from the end of the trajectory, the estimate at step t is

$$V_t^{\text{DR}} = \hat{V}(s_t) + \rho_t \left(r_t + \gamma V_{t+1}^{\text{DR}} - \hat{Q}(s_t, a_t) \right), \quad (5)$$

where $\rho_t = \pi_e(a_t|s_t)/\pi_\beta(a_t|s_t)$ and $\hat{Q}, \hat{V}$ are the value model’s predictions. The model supplies the baseline and the importance-weighted term corrects it by the observed residual, so where the weights collapse the estimate falls back to the model, and where one weight is large the residual of a single patient dominates.

Fitted Q Evaluation (FQE). FQE (Le Paine et al., 2020) avoids importance weights entirely. It trains a second Q-network to predict the return of the one fixed policy being evaluated, and not to find good actions. The network is fitted by repeated regression, each round using the previous round’s estimate as

its target:

$$Q^{(k+1)}(s, a) \leftarrow r + \gamma(1 - d) \sum_{a'} \pi_e(a'|s') Q^{(k)}(s', a'). \quad (6)$$

Each round therefore pushes the estimate one step further into the future. After K rounds, a state’s value is $V^{(K)}(s) = \sum_a \pi_e(a|s) Q^{(K)}(s, a)$, and the policy’s score is that value averaged over the evaluation split’s initial states. Because nothing divides by π_β , FQE does not decay as trajectories lengthen, and depends instead on the second network fitting well.

Because FQE is our primary estimator, we do not take its output on trust but validate it against a policy whose value we already know: the behaviour policy π_β itself. Running the same procedure with $\pi_e = \pi_\beta$ yields $\hat{V}_{\text{FQE}}(\pi_\beta)$, which we compare to the clinician’s empirical mean discounted return. A faithful estimator should recover that value. A large gap means the estimator is biased on this problem, in which case policies must be compared relative to $\hat{V}_{\text{FQE}}(\pi_\beta)$ under the same estimator, not against the empirical return (Sec. 5.1). We additionally re-run FQE for every policy under several estimator seeds, which separates the estimator’s own training noise from policy-training variance.

3.6 Measuring Data Support

Every estimator above answers the same question, what a policy is worth, and every one of them needs an additional assessment to be judged adequately. A policy that recommends treatments clinicians rarely or never gave forces the estimator to extrapolate, and that value cannot be trusted. The data-support measures in this thesis are built to detect that failure. We measure, in two independent ways, how well each policy’s greedy recommendations $a^* = \arg \max_a Q(s, a)$ are supported in the data on the test split. The first, **the propensity measure**, is the behaviour-policy propensity $\pi_\beta(a^*|s)$, the same variable that is in the importance-sampling denominator. The two differ in which action they evaluate. Importance sampling divides by π_β at the logged action, whereas this measure evaluates it at the recommended one. For a deterministic policy the importance weight is $1/\pi_\beta(a^*|s)$ where the clinician happened to take a^* and zero everywhere else, so the same scarcity that leaves an importance-sampling estimate undefined is reported directly by this measure, which is why it still applies to fitted-Q evaluation, where no importance weights exist. The second, **the k -NN measure**, is model-free: for each decision we take the nearest clinician states under standardised distance and record the fraction that took a^* , which we call that recommendation’s k -NN support. The two are deliberately independent, since the k -NN measure never consults the behaviour model. Agreement between them therefore cannot be an artefact of that model. A recommendation with low propensity or near-zero k -NN support lies outside the data, and we read its FQE value with that caveat in mind.

Neither ingredient is new. Propensity floors are standard practice in the diagnostics of importance sampling, and nearest-neighbour density estimates

are long established. The thesis claims their assembly into a per-policy trust measure read alongside the value estimate, and not the components.

4 Experimental Setup

Sec. 3 set out what each component is and why it was chosen. This section fixes the settings those choices were run at, and the protocol that we used.

Data and splits. Missing values take the most recent prior observation, then the training-set median. All features are z-scored on training statistics and clipped at $\pm 5\sigma$.

Stays are partitioned 70/15/15 by subject, giving 176,012 training, 38,011 validation and 37,874 test transitions from 25,442 ICU stays. The split is drawn once and reused for every model and variant in this thesis, so all reported comparisons are on identical data. Validation supplies the early-stopping diagnostic and selects the IQL actor, while test is read only for the final reported values.

Variant settings. CQL runs at a conservatism coefficient of $\alpha = 2.0$ and IQL at an expectile of $\tau = 0.7$, with its actor extraction at $\beta = 3.0$, pre-specified and not swept. Both values were fixed before this cohort was built and are not tuned against the results reported here, which keeps the comparison free of selection on its own outcome and means that neither is claimed to be the best setting for this cohort.

Evaluation protocol. Every estimator scores the policy that would actually be deployed, the greedy $\arg \max_a Q(s, a)$, and this is the same policy that the action distributions and the data-support measurements describe. FQE runs for 20 outer iterations of 3 inner epochs, batch size 512, Adam at $\text{lr} = 10^{-3}$, and is fitted separately for every policy and split.

Checkpoints and controls. Every method is evaluated at two checkpoints, the network after one epoch and the network after forty, and both are reported throughout. For IQL the two checkpoints vary the critic the actor is extracted against, and the actor itself is selected on validation loss in both. The reason is given with the result (Sec. 5.2). Three reference policies are scored under the identical procedure. The estimated behaviour policy is the comparator for improvement over observed practice. Behaviour cloning, a supervised classifier of the clinician’s action that applies no value learning at all, marks what imitation alone achieves. A never-treat control withholds both treatments at every step.

Experiments. Three experiments follow, all on the test split unless stated. In the first, every learned policy at both checkpoints is scored by all four estimators, and each estimator is judged on whether it is defined and on what its estimate rests on: effective sample size and per-step support for the importance-sampling family, the reward bound for DR, and refits at five evaluator seeds for FQE. In the second, each policy’s greedy recommendations are measured by both data-support measures, with the propensity measure’s floor at 0.01 and the k -NN measure taking the 50 nearest neighbours from 80,000 training decisions for each of 15,000 test decisions. In the third, the FQE ranking from the first

experiment is set beside the coverage ranking from the second, with within-stay differences against the behaviour reference.

5 Results

The central question is whether a reported advantage belongs to the policy or to the estimator that scored it. To investigate this, we take the estimators first and ask which of the four remain usable on this data and what their output can support a claim about. Every policy is then placed on whichever estimator passes it, and we close by measuring the coverage behind each of those values, a measurement that changes what the comparison means.

5.1 Estimator Reliability

Fig. 1 summarises the verdict on all four estimators across the thirty test fits, and the paragraphs below unpack these findings.

Trajectory-level weighting. Weighted importance sampling requires a logged trajectory whose actions match the evaluated policy at every decision, and a deterministic policy makes that rare. Averaged over seeds, 74 of the 3,817 test trajectories qualify for CQL at the early checkpoint and 76 for IQL, both around 2%. For D3QN the count falls to 6 at the early checkpoint and to none at all at the final one, where the estimate does not exist for any seed. Where WIS is defined its effective sample size is small enough to make the point on its own: a median of 6.9 equally weighted trajectories for CQL and 5.2 for IQL, and 1.0 for D3QN. 24/30 fits return a value, and none of them rests on enough data to separate policies.

Per-decision weighting avoids that product collapse, but it does not restore the support needed for a reliable estimate. By normalising at each step, a single mismatch no longer eliminates an entire trajectory. However, the estimator still depends on matched actions at every decision point, and these matches reach zero well before the trajectories end. Across the test split, CQL holds 918 of the 3,817 trajectories at decision 0, 30 by decision 10 and 2 by decision 15, after which none remain while 252 stays are still active. D3QN holds 180 at decision 0 and none from decision 11, where 1,249 stays are still active. Fig. 2 shows this fall for each method against the stays still in the ICU.

We therefore report an estimate only where every step in its sum retains at least one matched trajectory. Without that condition the estimator still yields a number: an unsupported step contributes zero through denominator clipping, and since all reward here is terminal, a zero at a late step removes the outcome from the estimate rather than marking it as missing, so what comes out is the value of a silently truncated horizon. Under this criterion per-decision weighting reports in 2 of the 30 test fits, with 22 of the 28 blanks withheld by the criterion rather than undefined. On validation, a split of near-identical size, it reports 19 of 30, so its availability is not a property of the policies alone but of which held-out

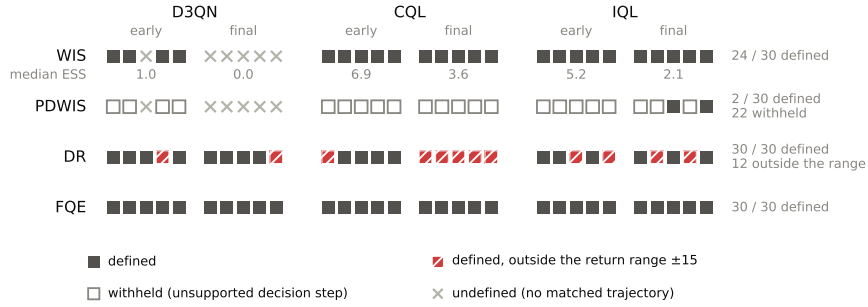


Fig. 1. Availability of each estimator on the test split, one mark per training seed for each method and checkpoint. A filled mark is a defined estimate, an open mark one withheld because a decision step retained no matched trajectory, a cross an estimate that does not exist because no trajectory matched the policy throughout, and a hatched mark an estimate outside the ± 15 the reward permits.

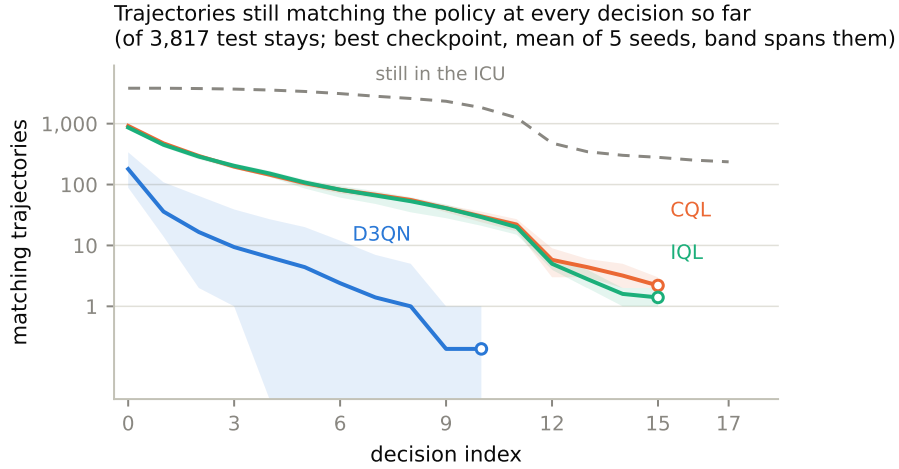


Fig. 2. The horizontal axis is the decision index within a stay, counted from zero at each stay’s first decision. The vertical axis is how many of the 3,817 test trajectories have logged actions matching the policy’s greedy choice at every decision up to that index, as a mean over five seeds with the band spanning them. The grey line represents the stays still in the ICU. Each line stops at its last non-zero count, since a log axis cannot show zero.

patients happen to be in hand. A study reporting a per-decision estimate on one split can therefore find none at all on another drawn from the same population.

Doubly robust estimation. DR fails differently from WIS. It does not degenerate, because its importance weights enter only the correction term (Jiang & Li, 2016), and it is defined on all 30 test fits, but the value cannot be used. Twelve of the thirty exceed the largest return the reward permits, the highest reaching 54.51 against a bound of 15. The spread within one configuration is as wide as the gap between methods: across five seeds of the same CQL run at the final checkpoint the estimate goes from 27.4 to 54.5, every one of them impossible.

The correction term is the reason why. Where support persists at a late step it often rests on a single trajectory, which self-normalisation then gives the full weight, so the term becomes one patient’s residual $r - \hat{Q}$. With a terminal reward of ± 15 and a baseline near 12, one death moves the estimate by roughly 27. Capping the weights at the percentile per-decision weighting uses changes nothing, leaving the same twelve fits out of bounds, so the dispersion is not an artefact of heavy tails. DR also does not reduce to its baseline: across the same fits it differs from the FQE value by as much as 41.95, and only nine of thirty sit within 1 of it. WIS and DR are limited by the same shortage of matched trajectories. WIS is left undefined, while the value estimate DR gives hinges on whether one patient lived or died. DR can produce policy-value estimates where WIS cannot, but it does not provide a reliable policy ranking on this dataset.

Fitted-Q evaluation. FQE needs no importance weights. Following Le Paine et al. (2020), we train a separate Q-network to evaluate each policy, using 20 outer Bellman iterations on the training transitions, and report $\hat{V}_{\text{FQE}} = \mathbb{E}_{s_0}[V^\pi(s_0)]$ on the held-out initial states. It is defined on all 30 test fits. That completeness is the whole of its advantage. FQE bootstraps through the actions the evaluated policy recommends, and where those actions fall outside the support of the behaviour policy it extrapolates and reports a value regardless, and nothing in its output separates that case from a well-covered one. The importance-sampling estimators are undefined under exactly those conditions. FQE is defined everywhere, so its output carries no signal about the support behind an estimate, which Sec. 5.3 therefore measures separately.

Levels and differences. Refitting FQE at five evaluator seeds, with the policy held fixed, separates the estimator’s own variability from the policy’s. Absolute levels move by a standard deviation of 0.11 to 0.22 depending on the policy, which is comparable to the margins the comparison turns on. Paired differences against the reference are steadier, because the errors of separate fits are correlated at 0.999. The exception is D3QN, where the difference varies by more than the level, since pairing reduces variance only between policies that visit the same states and D3QN does not visit the reference’s. Coverage therefore governs whether a margin can be measured at all, before any question of what the margin means.

What the estimates can support. Absolute FQE levels carry no external calibration, since no ground-truth value exists for any policy here. Each policy is therefore read against the estimated behaviour policy under the same estimator on the same held-out initial states, and only the ordering and the sign of a

within-stay difference are claimed. Whether that ordering reflects real differences between the policies or extrapolation over actions the data does not support is addressed in Sec. 5.3.

5.2 The Estimator’s Ranking

Under the protocol just established, every method is placed on the same comparison: fitted-Q evaluation, the greedy policy, the same held-out initial states, and two references. The first is the estimated behaviour policy, so a margin against it measures improvement over observed practice as the estimator models it. The second withholds treatment at every step, a cruder floor a policy must clear to beat doing nothing at all. Both are scored under the identical procedure, so the estimator’s shared bias cancels in a within-stay difference even though the absolute levels carry no external calibration. Differences are tested with the training seed as the unit of analysis and adjusted by Holm within a declared family of six, three methods against two checkpoint rules, separately for each reference.

No offline criterion selects a checkpoint. Both ends of training are reported because nothing available offline chooses between them. Validation loss is minimised at epoch 1 in all fifteen training runs, and rises afterwards, which Fig. 4 in the appendix shows for each run. Therefore the standard early-stopping rule applied literally selects a network one epoch into training for every method. Two mechanisms produce that one outcome. D3QN’s greedy Q values climb from 14.1 to 35.5, well past the largest return the reward permits, so its loss rises because its value estimates diverge. CQL and IQL hold Q inside that bound, and their validation loss rises anyway. A validation Bellman residual measures whether a Q -network agrees with its own targets. Tang and Wiens (2021) describe the related trap of reusing one split for both selection and estimation. We therefore report both checkpoints throughout and treat the choice between them as unresolved. Table 1 reports the result.

The ranking. D3QN at the early checkpoint sits at the top of the table, ahead of the behaviour reference by 0.79 and of withholding treatment by 0.60, and both margins hold under adjustment. IQL at the same checkpoint follows, clearing the behaviour reference by 0.36. Against the harder reference IQL does not separate, and neither does any other policy in the table. Absolute values, which the margins are taken from, are in Table 3 in the appendix. Only one of the six learned policies beats doing nothing, which places the never-treat control well inside the range of what the RL models produce. The control also scores above the behaviour reference, by 0.19, so under this estimator giving no treatment outranks observed practice.

The negative control fails. Withholding both treatments from every patient at every step has no clinical reading under which it is preferable to what the clinicians did, however the estimator places it above observed practice all the same. The likely mechanism is confounding by indication, because treatment intensity in the logged trajectories tracks how sick the patient already is, and fitted-Q evaluation scores a policy on the states those trajectories visit. The

Table 1. FQE value on the test split, given as the margin over the estimated behaviour policy so that every row is in one unit, where $\hat{V}(\pi)$ is the fitted-Q estimate of a policy’s value and π_β the estimated behaviour policy. Entries carry $\pm$ the standard deviation across five training seeds. The reference policies, shown in *italic*, are single fits and carry no spread. Margins are taken within stay and adjusted by Holm. Bold marks the two arms that separate from the behaviour policy, D3QN early at $p_{\text{adj}} = 0.005$ and IQL early at 0.009. No other method separates from it, and against the withhold control only D3QN early does, at 0.016.

Policy	$\hat{V}(\pi) - \hat{V}(\pi_\beta)$
D3QN (early)	+0.79 $\pm$ 0.20
IQL (early)	+0.36 $\pm$ 0.11
<i>Withhold treatment</i>	+0.19
CQL (final)	+0.02 $\pm$ 0.16
<i>Behaviour policy π_β (FQE reference)</i>	0.00
IQL (final)	−0.05 $\pm$ 0.35
CQL (early)	−0.05 $\pm$ 0.16
<i>Behaviour cloning (imitation floor)</i>	−0.23
D3QN (final)	−0.82 $\pm$ 0.72

control was included as a negative control, on the expectation that a sound estimator would rank it last. Its position instead shows that a value estimate built from these trajectories does not track clinical quality, and that holds for any such estimate, not only for fitted-Q evaluation.

Conservatism does not distinguish itself. CQL sits within 0.05 of the behaviour reference at both checkpoints, with adjusted p of 1.000 in each case. It neither improves on observed practice nor falls below it, and its margin against withholding treatment is negative at both ends of training. The method that most directly penalises out-of-distribution value inflation produces the policy least distinguishable from the reference it was measured against.

The checkpoint moves a method further than the method moves itself. D3QN spans +0.79 to −0.82 between epoch 1 and epoch 40, which is wider than the gap between the best and the worst policies at any fixed checkpoint. At epoch 40 it falls below the imitation floor and to the bottom of the table. Since the validation criterion selects epoch 1 for every method, the D3QN checkpoint it picks out is also the one that scores highest overall, and that same checkpoint is a network trained for a single epoch.

What the intervals do not carry. The seed standard deviations in Table 1 and the adjusted p values are both taken over training seeds, so they describe how much a policy moves when it is retrained. Refitting the estimator alone, at a fixed policy, moves D3QN’s within-stay difference by 0.30, against 0.20 for the same difference across training seeds. For D3QN (early checkpoint), refitting FQE contributes more variation than retraining does, and neither the interval nor the p value beside it includes that. CQL and IQL are steadier under refitting,

Table 2. Data support of each policy’s greedy recommendations on the test split, early checkpoint. “% unsupp.” covers all 37,874 test decisions and is the share of recommendations whose probability $\pi_\beta(a^*|s)$ under the estimated behaviour policy falls below 0.01. The two k -NN columns are computed on a random sample of 15,000 of those decisions, each compared against 80,000 training decisions with $k = 50$. k -NN support is the fraction of a decision’s 50 nearest clinician decisions that chose the recommended action, and “% zero k -NN” the share with no supporting neighbour at all. Bold marks the best and the worst entry in each column.

Policy	% unsupp.	k -NN support	% zero k -NN
Clinician (observed)	2.94%	0.187	6.53%
CQL ($\alpha = 2.0$)	0.00%	0.226	2.84%
IQL ($\tau = 0.7$)	0.07%	0.224	3.04%
D3QN	57.15%	0.069	50.15%

at 0.06 and 0.11, for the reason Sec. 5.1 gives: pairing reduces variance only between two policies that visit the same states.

The sign of a difference here is a statement about an estimator’s ordering under a common procedure. It is not an effect size, and it is not a claim about clinical value. Whether the ordering holds against the data it was computed from is the subject of the next section.

5.3 Value Against Data Support

The comparison of the previous section rests on FQE alone, which extrapolates wherever the evaluated policy recommends OOD actions. A check on it has to avoid the FQE regression altogether. Two measures do. The propensity measure reads each recommended action’s probability under the estimated behaviour policy and counts the share falling below 0.01. The k -NN measure fixes a state, takes the 50 nearest clinician decisions in the training data, and reports the fraction that chose the recommended action. Neither touches the FQE regression, the reward, or the outcome label. The k -NN measure uses no model and no threshold, since a recommendation supported by none of its neighbours is a fact about the data, so it carries the argument and the propensity measure corroborates it. Table 2 reports both measures.

D3QN, which fitted-Q evaluation ranks first, is the worst supported policy in every column (Fig. 3). Of its recommendations, 57% fall below the propensity floor and 50% are actions that none of the 50 nearest clinician decisions took. The two conservative methods place almost nothing outside support. Rank correlation between value and support across the four policies is -0.80 . Four policies are too few to test a correlation, the number is given to describe the ordering.

D3QN’s recommendations. Clinicians give a vasopressor in 26.5% of decisions. D3QN at the early checkpoint recommends one in 71% and agrees with the clinician on 7% of decisions, against 42% for CQL and IQL. Fig. 5 in the appendix shows where in the episode those recommendations fall. The same

Estimated value and data support, test split

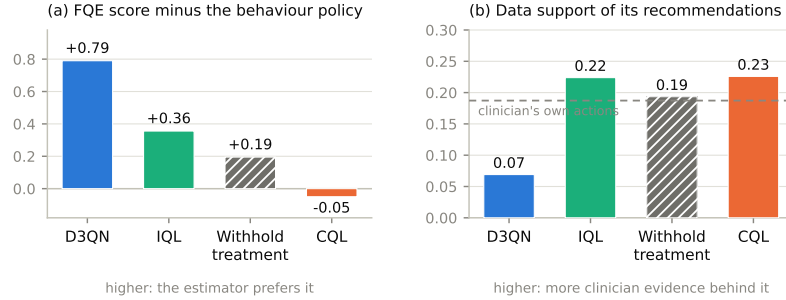


Fig. 3. Estimated value beside data support, test split, best checkpoint, mean over five seeds. (a) FQE value minus the estimated behaviour policy’s 12.52, the margin tested in Table 1. (b) Mean k -NN support as defined in Table 2, with the clinician’s own actions as the dashed reference. Seed standard deviations reach 0.20 on value and 0.04 on support.

training drives D3QN’s greedy Q values past the largest return the reward permits. Both networks bootstrap through a maximum over all twenty-five actions, most of them unsupported in any given state, and Double Q -learning does not prevent this. Decoupling selection from evaluation removes the maximisation bias of a single network, and two networks trained on the same transitions err in the same direction on actions neither has seen.

The final checkpoint. Training the same network to epoch 40 makes both quantities worse together. The vasopressor rate reaches 97%, the unsupported share 86%, and agreement with the clinician falls to 1%. FQE follows, dropping to 11.70, the lowest value in the comparison. So the estimator tracks the collapse once almost none of the policy’s recommendations are supported. At the early checkpoint, where the shift is already large, FQE rewards it instead.

Conservatism and coverage. CQL’s recommendations are better supported than the clinician’s own observed actions, at 0.00% unsupported against 2.94%, and 2.84% with no supporting neighbour against 6.53%, and the two measures agree on that ordering. CQL nonetheless sits within 0.05 of the behaviour reference at both checkpoints and below the never-treat control at both.

A clinical reading. The coverage result also has a clinical interpretation. Vasopressors are reserved for hypotension that persists after fluid resuscitation, so a policy that recommends one at 71% of decisions, close to three times the observed rate, is prescribing them to patients in whom clinicians almost never do. A network with no vasopressor data in the milder states has nowhere to anchor a value there except the sicker patients who actually received those doses, and both coverage measures record exactly that displacement. The clinician’s own k -NN support of 0.187 deserves the same reading. A clinician’s action agrees with fewer than one in five of the most similar decisions elsewhere in the data, and no single

action accounts for more than 17% of decisions, so observed practice is itself widely dispersed and a support of 0.226 counts as high against that background. The dispersion also explains how a policy comes to be better supported than the clinician without gaining anything. CQL favours actions that are common among comparable decisions, which raises coverage by construction.

One policy clears both measurements. IQL at the early checkpoint scores 12.87, above the behaviour reference by 0.36 with all five training seeds on the same side of it, and its recommendations carry a k -NN support of 0.224 against the clinician’s own 0.187. It is the only policy the estimator places above the reference and the data-support. Three qualifications bound what that can be taken for. The margin does not separate IQL from the never-treat control, where the adjusted p is 0.125. It is small beside the machinery producing it, since the reference itself moves by 0.13 across independent refits of the estimator. It also belongs to one of two checkpoints, selected by a criterion that does not discriminate between them, and the other checkpoint of the same method sits below the reference. The two measurements agree on IQL early, and that agreement is not evidence the policy would help a patient.

The two measurements do not order the policies the same way. Read alone, FQE places D3QN first, and D3QN’s recommendations are the least supported in the data. k -NN support places CQL first, and CQL does not separate from the never-treat control. Neither ranking can be inferred from the other. Read together they do not identify every failure. The never-treat control recommends the action clinicians take at 16.6% of decisions and carries a k -NN support of 0.19, matching the clinician’s own. The value estimate places it above observed practice. Both measurements pass a policy that no clinician would defend.

6 Discussion

The central finding is a divergence between two measurements of the same policies, and neither ranking can be recovered from the other. A value estimate read alone can rank first a policy the data does not support, and a coverage measure read alone can rank first a policy no better than withholding treatment. Sec. 5.3 shows both happening. Since single-estimator reporting remains common in this literature, we take the pairing to be the more transferable of our findings.

FQE and the evaluated policy depend on the same state-action pairs, and where those pairs are sparsely covered both extrapolate. A policy that recommends a treatment clinicians rarely gave in a given state is scored by a network with no observations of that treatment in that state either. That violates the full support assumption OPE analysis relies on for unbiasedness (Voloshin et al., 2021), and FQE produces a finite estimate all the same. The train and test split does not correct this. A patient’s record contains an outcome only for the treatment they were given, so held-out patients supply no observations of a treatment that was rarely given to anyone.

Poor coverage has a second consequence. Each policy is scored by its own fitted-Q network, and the errors of two such fits are strongly correlated across

evaluator seeds, so the paired difference has lower variance than either absolute level. That correlation requires both policies to visit the same states. D3QN does not meet that condition, and refitting moves the difference more than the level.

On the RL side the comparison locates the failure precisely. Levine et al. (2020) predict distributional shift in theory, and here it is visible in the treatment recommendations themselves, with an unconstrained agent (D3QN) driving its vasopressor rate to 71% of decisions against a clinical rate of 26.5%. The two forms of conservatism separate here. CQL penalises out-of-distribution Q -values explicitly, which kept its recommendations within support at both checkpoints and produced a policy indistinguishable from the reference at both. IQL obtains the same restraint implicitly, by never querying an unsupported action in its Bellman target, and at the early checkpoint it holds comparable coverage while scoring above the reference. Recommending supported actions does not by itself cost a policy its margin. That contrast rests on one setting of α and one of τ , neither swept, so it locates a question and does not settle it. We would expect methods that constrain the action set directly, such as BCQ, to behave like CQL on the coverage axis, a test we did not run.

Pairing the two measurements is therefore necessary on this data, and it is still not sufficient, since the never-treat control passes both. Value and coverage together bound what can be believed about a policy that recommends unsupported actions, and they say nothing about a policy that recommends supported ones for the wrong reason. A third check has to come from outside both, from clinical review or from a model of why treatment was given in the first place.

The comparison these measurements support is correspondingly narrow. It is ordinal. Absolute levels carry no external calibration. Fitted-Q evaluation places the estimated behaviour policy at 12.518 against an empirical discounted return of 7.04 on the same patients, an overestimate of roughly three quarters. Every value is scored against a model of clinician behaviour rather than against observed outcomes, so a policy above the reference exceeds that model and not the clinicians themselves. The protocol supports the sign of a within-stay difference between two policies scored by one estimator on identical trajectories, and it says nothing about the size of a clinical effect. No policy described here was ever run on a patient, and only a prospective trial could settle whether a better score means a better outcome.

Three features of the study limit how far these results generalise. The first is the cohort, a first-episode Sepsis-3 population of 25,442 stays with 25.0% ninety-day mortality taken from the official MIMIC-IV derivation. It comes from a single US academic centre, and as Nauka et al. (2025) show for ICU-to-emergency transfer, transportability across settings is not validated here. Repeating the comparison on a second source such as eICU would test whether the ordering is a property of the methods or of this cohort.

The second is partial observability. The state is a fixed four-hour snapshot fed to a feed-forward network, so we assume the patient is Markov and fully observed, which is not quite true. Trends over time go unmodelled, as does the irregular, partial sampling of ICU data, and carry-forward imputation replaces

those gaps with the last observed value. The state omits cumulative fluid balance, renal replacement therapy, and sedation, three variables that would each help distinguish a genuine change in the patient’s condition from an artefact of unmeasured treatment. The formulation is therefore a partially observable Markov decision process modelled as a fully observed one, carrying no observation function and no belief state. Much of what distinguishes one patient from another never reaches the dataset either, so two patients with the same recorded state may still differ in ways that matter. The previous interval’s treatment levels are set to zero at each episode’s first decision, because no earlier interval exists to carry forward, so a patient already receiving norepinephrine when the window opens is presented to the network as untreated. That affects 3,817 of the 37,874 test decisions, one in ten, and it concentrates at onset, where 68% of decisions are some episode’s first. Fitted-Q evaluation scores every trajectory from its first decision, so this may affect its results.

The third is offline model selection, and it is the main open question left for future work. Validation loss selects epoch 1 for every method here, and the policies at epoch 1 and epoch 40 differ by more than the methods differ from each other. Future work could select the checkpoint by estimated value among only those whose recommendations stay within the support of the behaviour policy.

7 Conclusion

This thesis asked how reliably offline reinforcement learning policies for sepsis treatment can be evaluated. The answer starts with the estimators, and three of the four tested return nothing a claim can rest on. Trajectory-level weighting reports on a handful of patients, per-decision weighting is withheld in all but two of thirty fits, and doubly robust estimation returns values no policy under this reward could reach. Only fitted-Q evaluation yields a usable estimate for all six policies. Its output supports a relative ranking of those policies under a common evaluation procedure but not an absolute value or an estimate of clinical effect.

Coverage varies sharply across policies, from almost none of CQL’s or IQL’s recommendations falling outside the support of the behaviour policy to more than half of D3QN’s.

Estimated value and data support do not identify the same policies. FQE ranks highest the policy the data supports least, and the method built to prevent that failure (CQL) keeps its recommendations well within support while producing a policy indistinguishable from the estimated behaviour policy and from withholding treatment entirely. A value estimate identifies neither case on its own. Read alongside k -NN support both become legible, and evaluating an offline policy on data of this kind requires at least the two together. The pair is not sufficient either, since the never-treat control passes both measurements. One policy satisfies both, IQL at its early checkpoint, which clears the behaviour reference by 0.36 and recommends actions better supported than the clinician’s own, and which does not separate from that same control.

One deeper limit remains, and it is in the objective rather than in the method. The reward here encodes ninety-day survival alone, so a policy is scored on mortality and on nothing else. That single term says nothing about what treatment costs a patient while it is given or about their quality of life afterwards. Vasopressors and intravenous fluids, the only two treatments in the action space, carry burdens of their own, along with the invasive monitoring, sedation and prolonged admission that typically accompany them. An intervention that improves the reward may not be worth its burden. A ratio of risk and cost to benefit would have to be taken into account. Furthermore, the patient's preferences and consent lie outside the formulation altogether. A policy learned this way describes one narrow sense of a better outcome that needs to be built upon.

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Appendix

A Supporting Figures

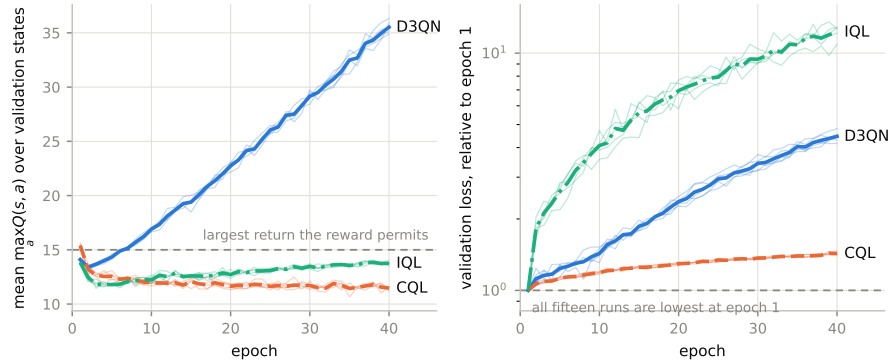


Fig. 4. Training behaviour over forty epochs, five seeds per method. Both panels are measured on a 10,000-transition sample of the validation split, the split the early-stopping criterion is computed on. Faint lines are individual seeds and the heavy line is their mean. Left: the mean of $\max_a Q(s, a)$, against the largest return the reward permits. D3QN crosses that bound early in training and reaches 35.5, while CQL and IQL stay below it throughout. Right: validation loss scaled by each method’s own epoch-1 value, on a log axis. The quantity behind that loss differs by method, so levels are not comparable across the three and only the shape of each curve against its own starting point is. All fifteen runs take their lowest validation loss at epoch 1.

Share of patients treated at each decision, test split
(early checkpoint, mean of 5 seeds, band spans the seeds)

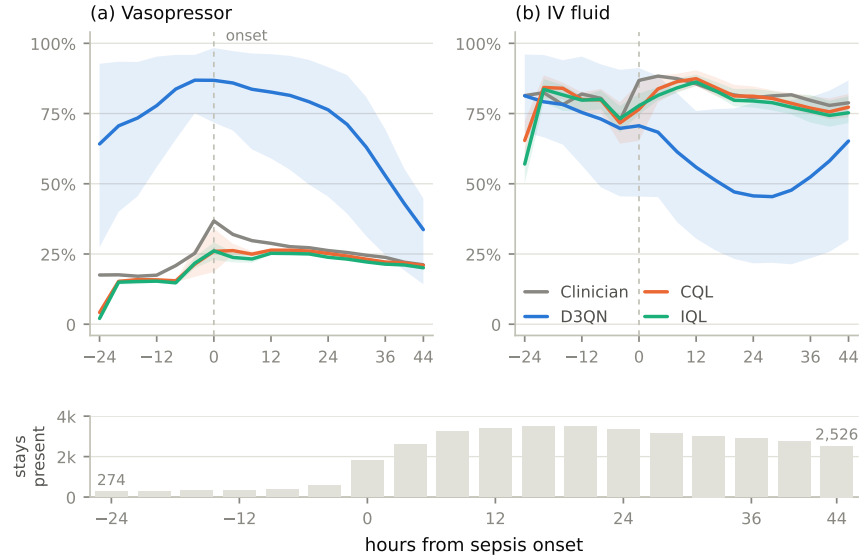


Fig. 5. Share of patients receiving each treatment at every decision in the episode window, test split, early checkpoint, mean over five seeds with the band spanning the seed minimum to maximum. The lower panel gives the denominator, which is not flat: episodes begin at ICU admission or 24 hours before onset, whichever is later, so the pre-onset bins hold only the minority admitted early.

B Per-Seed Results

Table 3. FQE value on the test split for every training seed, greedy policy. These are the absolute values behind Table 1, which reports each method as a margin over the estimated behaviour policy at 12.52. The final column’s standard deviation is the interval given there.

Method	Ckpt.	42	43	44	45	46	Mean $\pm$ SD
D3QN	early	12.974	13.286	13.384	13.403	13.495	13.308 $\pm$ 0.201
D3QN	final	12.436	10.492	11.803	11.863	11.897	11.698 $\pm$ 0.720
CQL	early	12.567	12.247	12.608	12.360	12.559	12.468 $\pm$ 0.157
CQL	final	12.552	12.445	12.363	12.535	12.791	12.537 $\pm$ 0.161
IQL	early	12.901	12.793	12.803	13.053	12.821	12.874 $\pm$ 0.109
IQL	final	12.294	12.818	12.111	12.869	12.256	12.470 $\pm$ 0.348